AI ASSISTED CODING - 22

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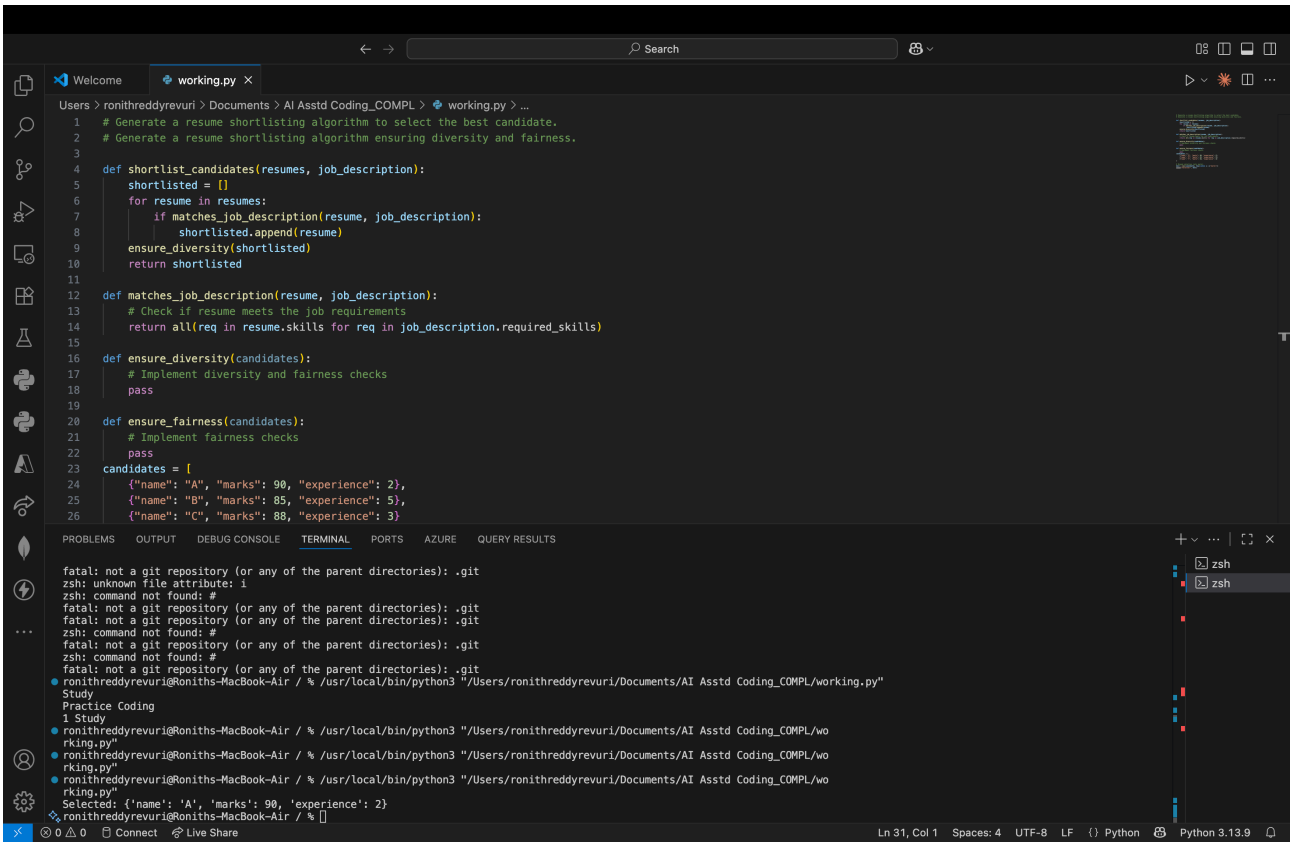
TASK1 : BIAS DETECTION IN AI
PROMPT :
GENERATE A RESUME SHORTLISTING ALGORITHM TO SELECT THE BEST CANDIDATE.

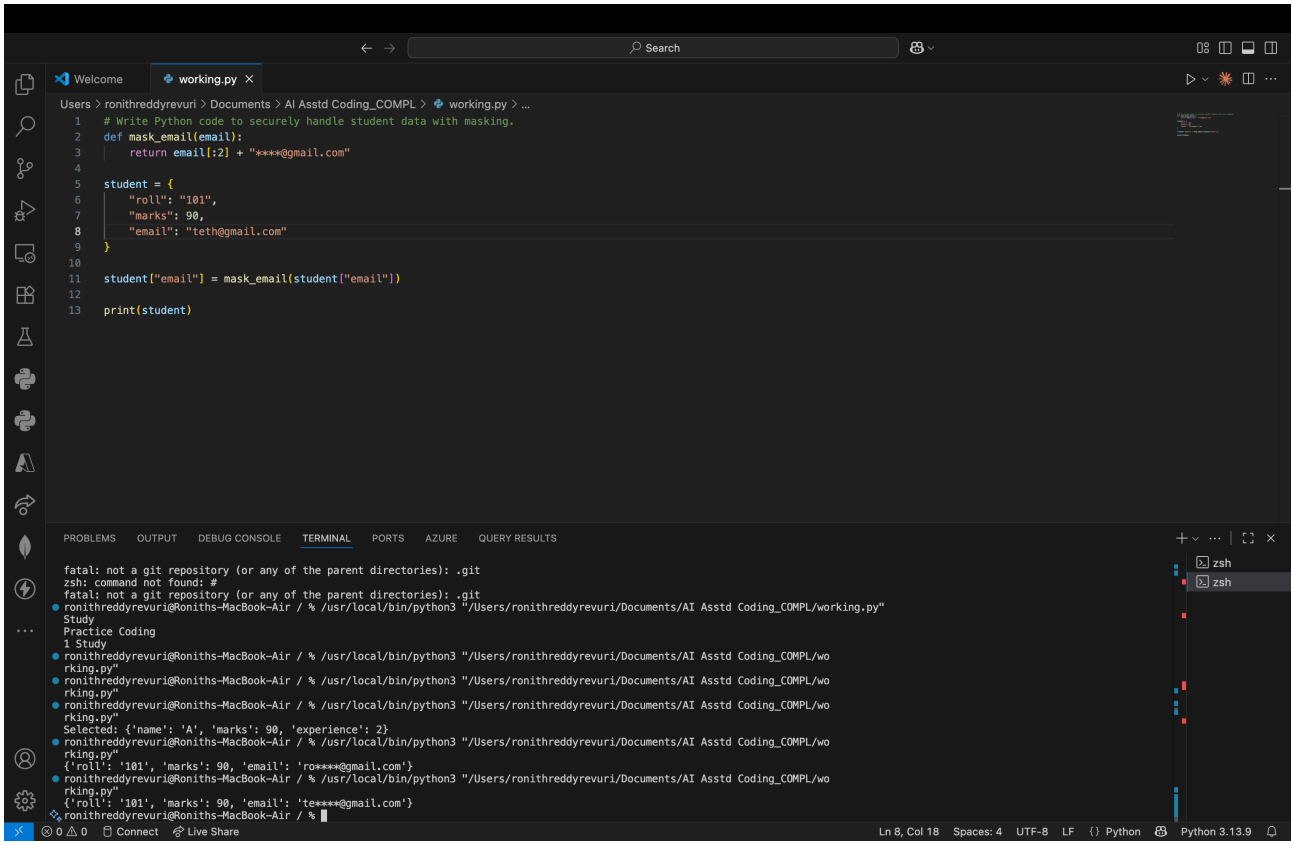
EXPLANATION :
WHEN WE ASK AI TO SELECT THE “BEST CANDIDATE”, IT MAY ONLY CHOOSE BASED ON MARKS OR EXPERIENCE. BUT WHEN WE ASK IT TO “ENSURE DIVERSITY”, IT TRIES TO CONSIDER FAIRNESS AND DIFFERENT BACKGROUNDS. THIS SHOWS AI RESULTS DEPEND ON HOW WE ASK, AND IT CAN BECOME BIASED.



TASK 2 – PRIVACY & SENSITIVE DATA

PROMPT :
WRITE PYTHON CODE TO SECURELY HANDLE STUDENT DATA WITH MASKING.

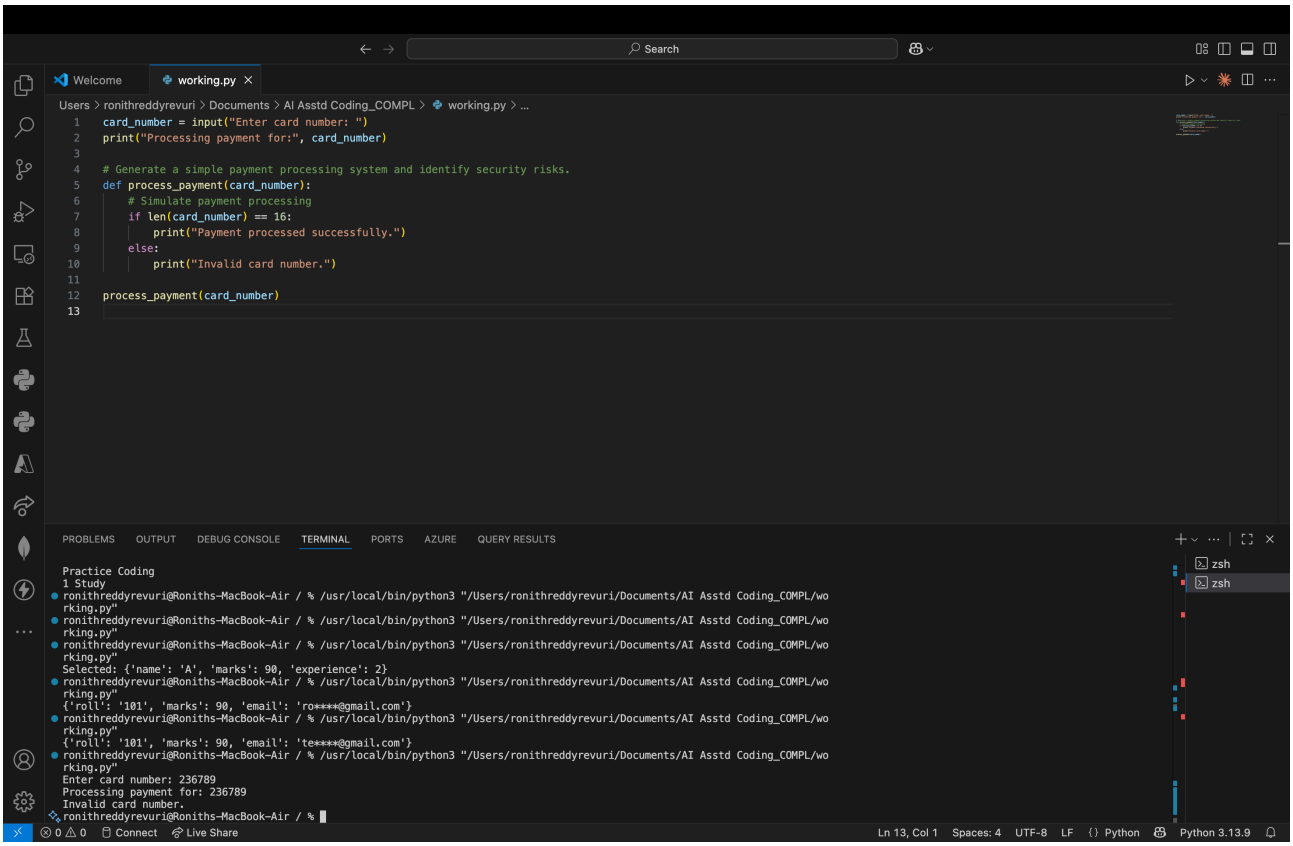
EXPLANATION :
SENSITIVE DATA LIKE EMAILS SHOULD BE HIDDEN OR MASKED. THIS PROTECTS USER PRIVACY. DEVELOPERS MUST FOLLOW ETHICAL RULES WHEN HANDLING PERSONAL DATA.



TASK 3 : AI SECURITY RISKS

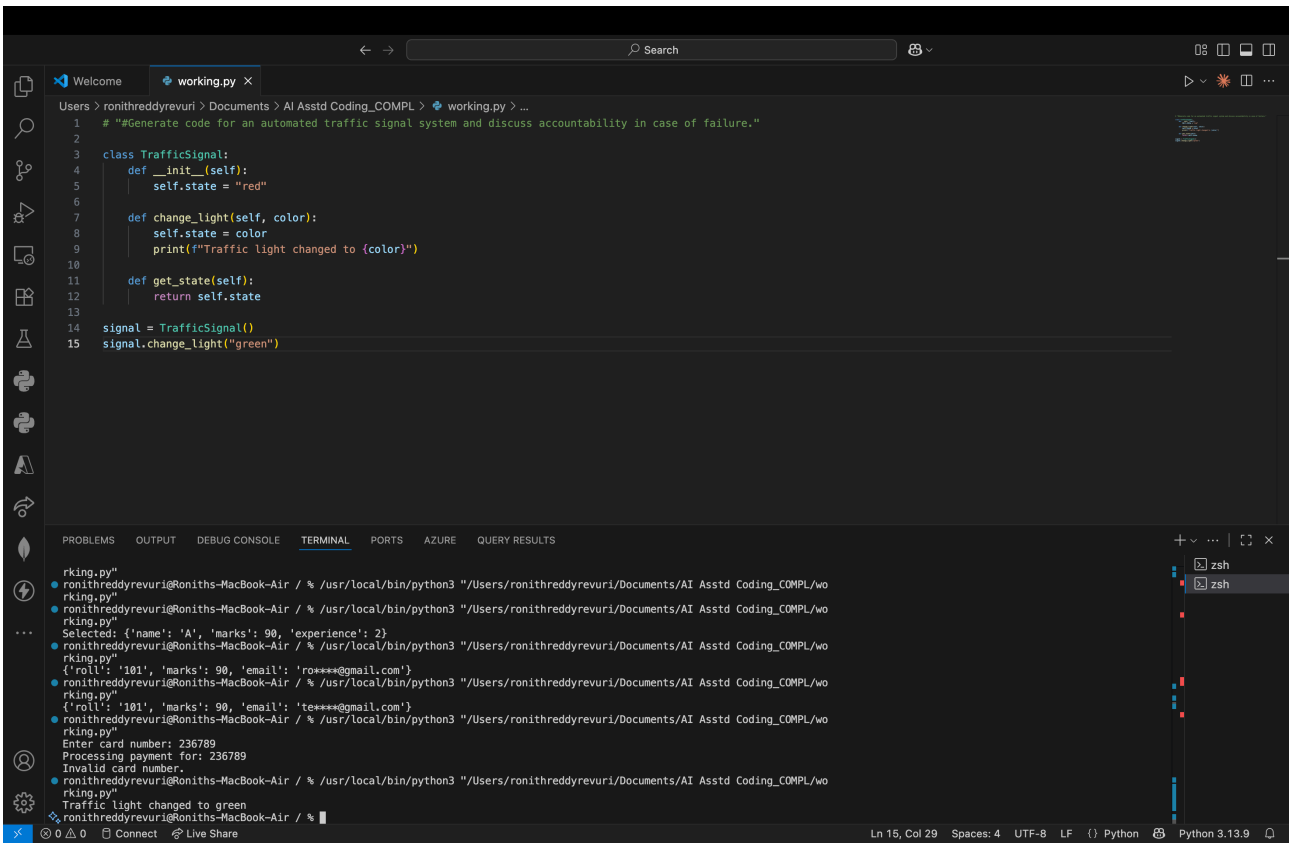
PROMPT :
GENERATE A SIMPLE PAYMENT PROCESSING SYSTEM AND IDENTIFY SECURITY RISKS.

EXPLANATION :
STORING OR PRINTING CARD DETAILS IS UNSAFE. INPUT VALIDATION AND HIDING DATA ARE IMPORTANT. SECURE CODING PREVENTS FRAUD AND MISUSE.



TASK 4 : ACCOUNTABILITY (THEORY)
PROMPT : # "#GENERATE CODE FOR AN AUTOMATED TRAFFIC SIGNAL SYSTEM AND DISCUSS ACCOUNTABILITY IN CASE OF FAILURE."

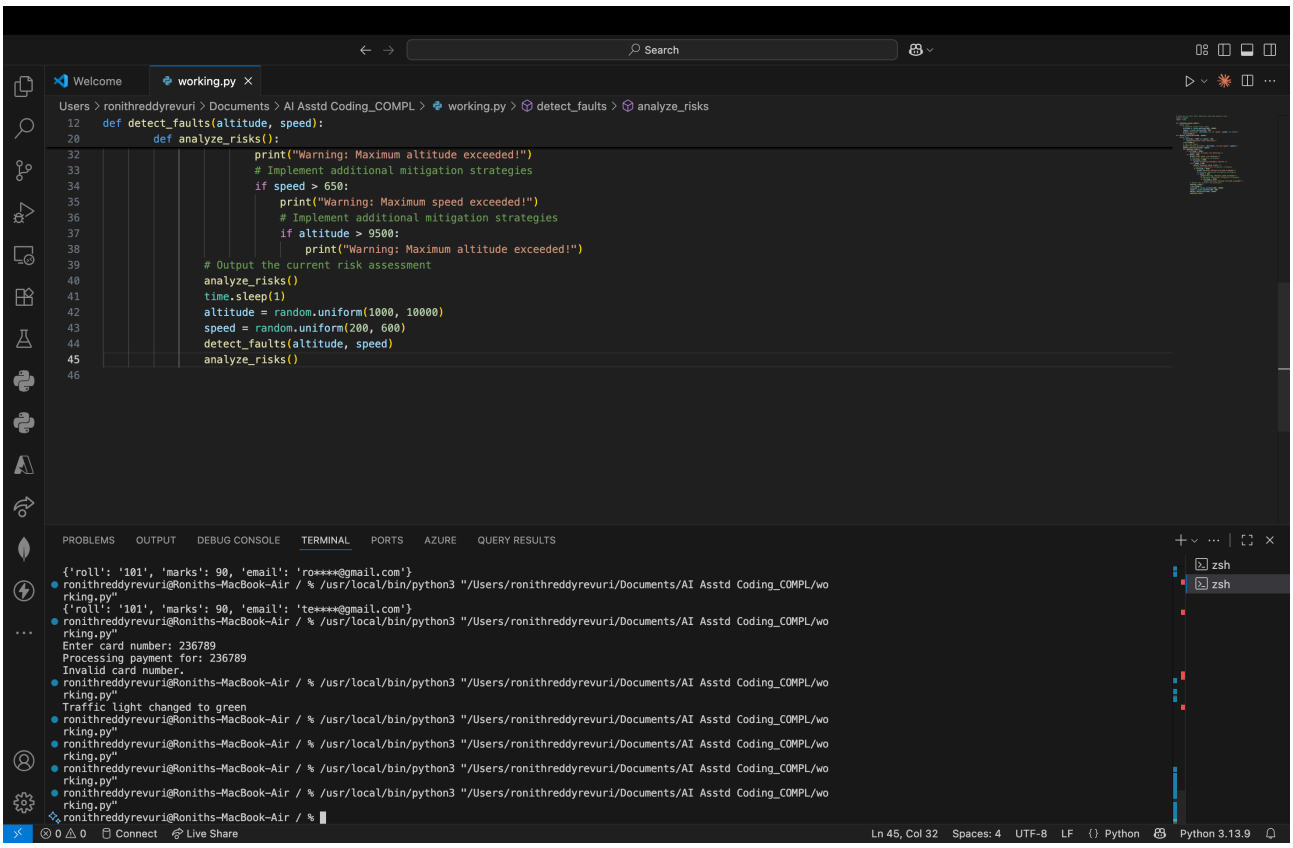
EXPLANATION :
IF AL-GENERATED SYSTEMS FAIL, RESPONSIBILITY CANNOT BE PLACED ONLY ON AL SOFTWARE ENGINEERS, DEVELOPERS, AND AUTHORITIES SHARE ACCOUNTABILITY. HUMAN OVERSIGHT IS ESSENTIAL TO ENSURE SAFETY AND RELIABILITY.



TASK 5 : LIMITATIONS OF AI

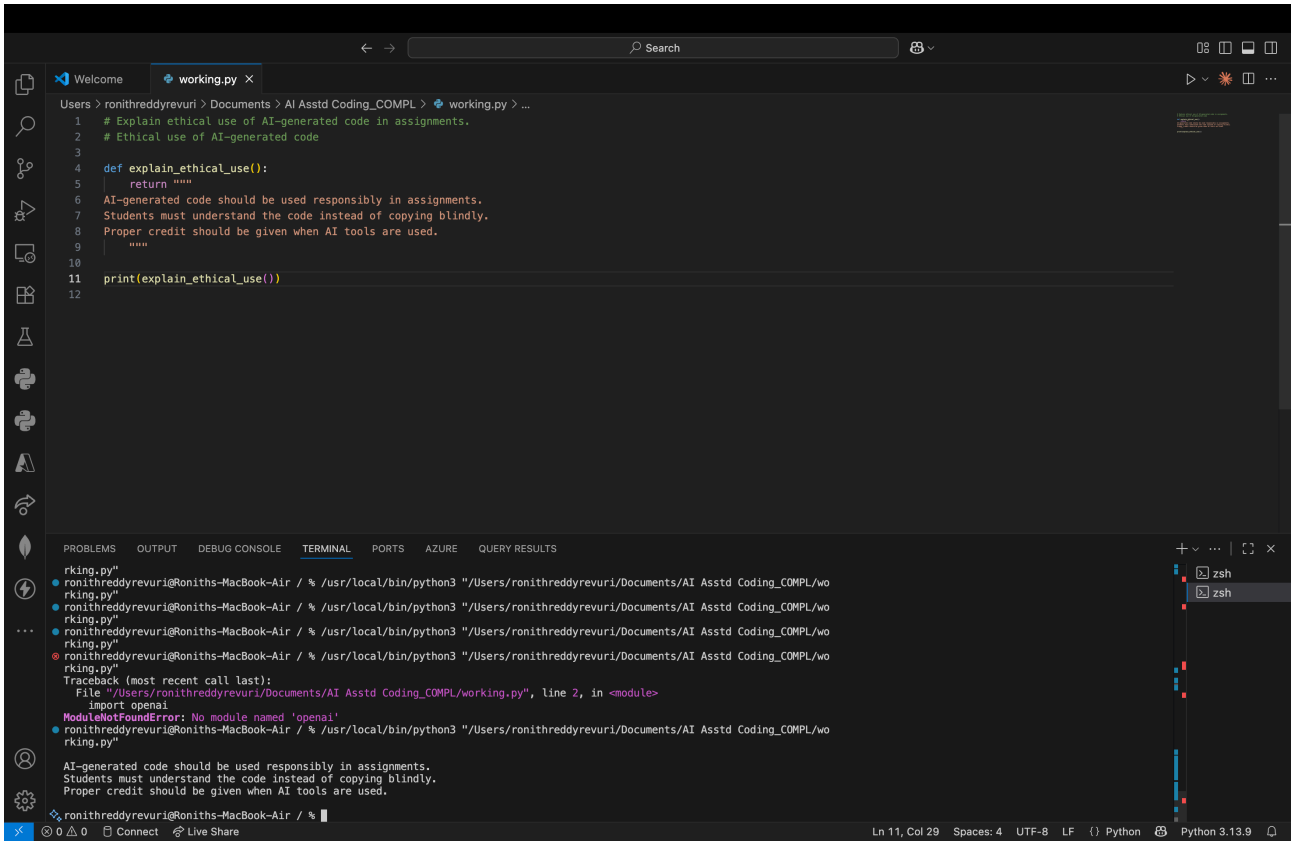
PROMPT :
GENERATE AIRCRAFT FAULT DETECTION CODE AND ANALYZE RISKS.

EXPLANATION :
AI-GENERATED CODE CANNOT BE FULLY TRUSTED IN CRITICAL SYSTEMS LIKE AVIATION. EXPERT VALIDATION AND STRICT TESTING ARE REQUIRED. HUMAN OVERSIGHT ENSURES SAFETY.



TASK 6 : ETHICAL USE OF AI CODE

PROMPT :
EXPLAIN ETHICAL USE OF AI-GENERATED CODE IN ASSIGNMENTS.

EXPLANATION :

USING AI IS HELPFUL BUT MUST BE DONE RESPONSIBLY. BLIND COPYING CAN LEAD TO MISTAKES AND PLAGIARISM. ETHICAL USE ENSURES LEARNING AND FAIRNESS.